



K. A. D. AVISHKA LAKSHAN KANAKASEKARA

Data Science Undergraduate | Data Analysis & Business Intelligence | Machine Learning
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PROFILE

Third-year Data Science undergraduate at SLIIT with a 3.82 GPA and three Dean's List awards (4.00 GPA in both Year 2 semesters). Experienced in building end-to-end data pipelines, designing dimensional data warehouses, and developing Power BI dashboards for business reporting. Applied machine learning experience through a statistical modelling research project comparing five predictive models with cross-validated evaluation. Strong foundation in mathematics, statistics, and software engineering, with a track record of leading technical project teams from design through deployment.

EDUCATION

- BSc (Hons) Information Technology — Specialisation: Data Science**2023 – Present
Sri Lanka Institute of Information Technology (SLIIT), Malabe | Currently Year 3
 - Cumulative GPA: 3.81 | Y1S1: 3.86 ★ | Y1S2: 3.38 | Y2S1: 4.00 ★ | Y2S2: 4.00 ★ | Dean's List — 3 semesters
- BSc (Physics, Computer Science, Mathematics)**2023 – Present
University of Peradeniya
- G.C.E. Advanced Level — Physical Science Stream**Completed 2021
Dharmaraja College, Kandy
 - Physics: B | Chemistry: B | Combined Mathematics: C | Z-Score: 1.3706

TECHNICAL SKILLS

- Programming Languages:** Python, R, SQL, Java, Kotlin, JavaScript, C++, C
- Data Science & Machine Learning:** Random Forest, Gradient Boosting, Decision Trees, Regression, Classification, Cross-Validation, Feature Engineering, Label Encoding, Model Evaluation (Confusion Matrix, F1, Precision, Recall)
- Data Warehousing & ETL:** SSIS, SSAS, Star/Snowflake Schema, SCD Type 2, Dimensional Modelling, OLAP, Staging Pipelines, SQL
- BI & Visualisation:** Power BI (DAX, Power Query, Power BI Service, Tooltips, Drill-Through), Excel (PivotTables), Tableau, Jupyter Notebook
- Databases:** MySQL, MS SQL Server, MongoDB
- Web Development:** Spring Boot, React.js, Node.js, Express.js, REST API, JWT, OAuth 2.0
- Tools & Platforms:** Git, GitHub Actions (CI/CD), Postman, Android Studio, VS Code , Jupyter notebook

KEY PROJECTS

- End-to-End Customer Churn Analysis Dashboard & Machine Learning Prediction System**2026
Individual Project | SQL Server, Power BI, Python, Scikit-learn
 - Developed SQL-based ETL pipelines, staging/production tables, and analytical views to clean, transform, and prepare telecom customer data for reporting and predictive modelling.
 - Designed interactive Power BI dashboards with DAX measures, Power Query transformations, drill-through insights, tooltips, and executive-level KPI visualisations for customer retention analysis.
 - Performed exploratory data analysis (EDA) to identify churn patterns across demographics, contracts, services, tenure, and geographic regions.
 - Trained and evaluated a Random Forest classification model achieving 84% accuracy, using feature engineering, label encoding, confusion matrix analysis, and feature importance evaluation.
 - Predicted 378 potential future churners and integrated model outputs into Power BI dashboards to support data-driven retention strategies.

Technologies: SQL Server, SSMS, Power BI, DAX, Power Query, Python, pandas, scikit-learn, Jupyter Notebook, Excel

Statistical Modelling Research: Data Maturity & Organisational Confidence

2025 – 2026

SLIIT — IT3040 | Team Leader | Kaggle-sourced dataset adapted to PwC & McKinsey frameworks

- Led a team of 4, analysing 1,000+ companies across 18 variables to quantify how data maturity drives organisational confidence
- Built a full analytics pipeline in R: Pearson correlation ($r = +0.954$), simple linear regression ($R^2 = 91\%$), one-way ANOVA ($F = 1059$, $p < 0.001$)
- Compared 5 predictive models (MLR, Ridge, Decision Tree, Random Forest, Gradient Boosting); the best model achieved 5-fold CV $R^2 = 0.917 \pm 0.008$ with zero overfitting
- Key finding: Level 5 firms averaged 99.3% confidence vs. 29.9% at Level 1 — a 69-point gap

Stack: R, Statistical Modelling, Regression, ANOVA, Cross-Validation, Predictive Analytics

Data Warehousing & Business Intelligence System

2025 – 2026

SLIIT — IT3021 | Kaggle-sourced academic records dataset

- Designed a production-grade star schema data warehouse with 1 fact table and 4 dimension tables (~30,000 fact rows), including SCD Type 2 historical tracking
- Built a 3-layer ETL pipeline (TXT, CSV, XLSX → Staging → DW) using modular SSIS packages with lookup and conditional split transformations
- Deployed an SSAS multidimensional cube and demonstrated all 5 OLAP operations via Excel PivotTables
- Published 4 Power BI reports to Power BI Service with custom DAX measures, including average mark, pass rate %, and total students

Stack: MS SQL Server, SSIS, SSAS, SSRS, Power BI, DAX, Star Schema, SCD Type 2

Smart Campus Operations Hub (UNI PULSE)

Apr – May 2026

SLIIT — IT3030 | Team Leader

- Led design and development of a full-stack platform with booking workflow, conflict detection, real-time notifications, Google OAuth 2.0, and role-based access control (User / Admin / Technician)
- Directed REST API design in Spring Boot, the React.js front end, and a GitHub Actions CI/CD pipeline end-to-end

Stack: Java, Spring Boot, React.js, MySQL, GitHub Actions, Google OAuth 2.0

Online Clothing Store Management System

Jul – Oct 2025

SLIIT — MERN Stack Group Project

- Built a full-stack e-commerce platform with a 3D cloth customiser (Three.js), inventory management, supplier management, and JWT-secured authentication

Stack: MongoDB, Express.js, React.js, Node.js, Three.js, JWT

HONOURS & AWARDS

- Dean's List — Year 2, Semester 2 (GPA: 4.00), SLIIT — May 2026
- Dean's List — Year 2, Semester 1 (GPA: 4.00), SLIIT — Jan 2026
- Dean's List — Year 1, Semester 1 (GPA: 3.86), SLIIT — Oct 2024
- Kids, Junior, Senior & Open Athletics Championships 2013, Kandy District Athletics Association — Oct 2023
- All Island Graded Examination — Spoken English (91/100), Institute of Western Music and Speech — Sep 2010

EXTRACURRICULAR ACTIVITIES

- Dharmaraja College, Kandy — Astronomy Club, Quiz Club, Nature Club: analytical thinking, teamwork, and problem-solving in academic and field competitions
- Sports: Athletics (Kandy District Championships), Carrom, Swimming, Cricket

REFERENCE

Dr. Prasanna Sumathipala

Email: prasanna.s@slit.lk

Dean, Matara Centre | Department of Information Technology, Faculty of Computing

Sri Lanka Institute of Information Technology (SLIIT)

Ph.D. — Sheffield Hallam University, UK | M.Sc. Computer Science — UCSC, University of Colombo | B.Sc. Computer Science — University of Colombo